DustLedger

Estimate each construction site's own dust — so inspectors can go where the evidence points

Team: ‹TEAM NAME› · **Team ID:** ‹ID› · **Theme 2:** Rising AQI Levels in Modern Urban Cities
‹Member 1 — role› · ‹Member 2 — role› · ‹Member 3 — role› · ‹Member 4 — role› · ‹email / phone›

Status today: data pipeline + replay BUILT · sites and sensors in the replay SIMULATED · sensor ring + misting PLANNED

This winter, Delhi's plan restricts dust-generating construction citywide

When (reported plan)	What
Nov 1 – Jan 31	dust-generating demolition and outdoor civil works **prohibited**
Dec 10 – Jan 20	**total ban** on construction and demolition
GRAP emergencies	stage-wise restrictions on top (some project categories exempt); GRAP-3 halted construction for 16 days in 2025 (₹10,000 compensation announced)
Every site	100% CCTV coverage mandated

Restrictions follow the activity and the pollution stage — not each site's measured dust.

Sources: Delhi Winter Action Plan 2026-27 as reported by The News Mill (Sept 2026; secondary, not the operative order); Hindustan Times (Nov–Dec 2025). Fact rows F1, F3.

A monitor can say the air is bad — not which site made it bad

- **City-scale sensing:** 47 reference monitors in Delhi (going to 60); the planning norm is one per 25-50 km²
- **Site compliance = checklists:** anti-smog guns (₹7,500 fine per missing gun per 5,000 m²), CCTV, self-audit uploads
- **Fines by area, not by dust:** CAQM's 2024 schedule, as reported, sets ₹1 lakh to ₹2 lakh per construction-site violation depending on area, doubled under GRAP (current operative order UNVERIFIED)
- **Trust is strained:** in Oct 2025 water tankers were filmed spraying at the Jahangirpuri reference monitor; CPCB says its data can't be manipulated

Penalties track checklists and area, not the dust a site actually releases.

Sources: Hindustan Times (2026; Feb 2024, reporting the CAQM order of 6 Feb 2024); Newslaundry (30 Oct 2025); New Indian Express (3 Dec 2025). Fact rows F5, F6, F25.

A small source, handled with blunt tools

2-5%

of Delhi's ambient PM10 came from construction material in IIT Kanpur's receptor study for DPCC (data 2013–14; track-out onto roads is counted as road dust).
Construction & demolition: 3.6% of PM10 *emissions*; concrete batching another 10%.

1,378,199

construction workers on Delhi's welfare-board rolls (Dec 2024) — only 89,518 active members.
In a rapid survey reported by Down To Earth, 5 of 75 site workers were registered; none of those had received the ₹8,000 aid.

Stops fall hardest on daily-wage workers — for a source that was a few per cent of city PM in that study.

Sources: IIT Kanpur (2016) for DPCC, §4.6 and Table 3.1 (primary; 3.6% is our arithmetic: 5.1668 of 143 t/d); Down To Earth (Dec 2024; survey by Aledia). Fact rows F7–F10.

Why today's tools fall short

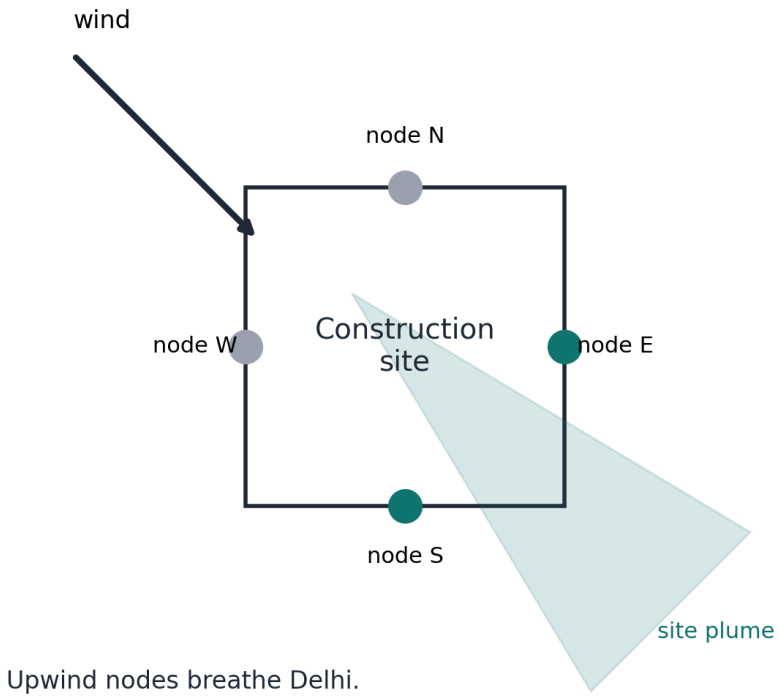
What exists	The gap
Reference monitors (CAAQMS)	city scale — can't single out a site
Compliance monitors streaming to the DPCC cloud (e.g. Airveda, Prana Air)	absolute PM; winter PM10 averaged 314 ± 77 µg/m³ across NCR sites in TERI–ARAI's 2016-17 campaign, so a site is a small difference on top
Anti-smog guns, sprinklers	run on timers — no feedback on whether dust fell
City AQI forecasts (IITM, Google)	tomorrow's city AQI, not today's dusty site
Fence-line dust systems (e.g. Aeroqual, Envirosuite)	established for mines and industry; we didn't find a low-cost workflow for Indian construction sites in the product pages we reviewed

Sources: research/landscape.md §3–4 (vendor pages; TERI–ARAI). Fact row F11.

DustLedger: a fence-line ring that subtracts Delhi

1. **Sense** — 4 PM nodes + 1 wind sensor on the boundary; solar + LoRa; LTE gateway **PLANNED**
2. **Estimate** — downwind minus upwind = our estimate of the site's own dust (an inference, not proof of cause); **"inconclusive" below 0.5 m/s wind** **SIMULATED**
3. **Act** — mist only when the estimate says the site is adding dust; flow-metered **PLANNED**
4. **Record** — signed, hash-chained records → a list that tells inspectors where to look first **PLANNED**

A check for confused or swirling wind, and for nodes that disagree with the wind, is planned — not built.

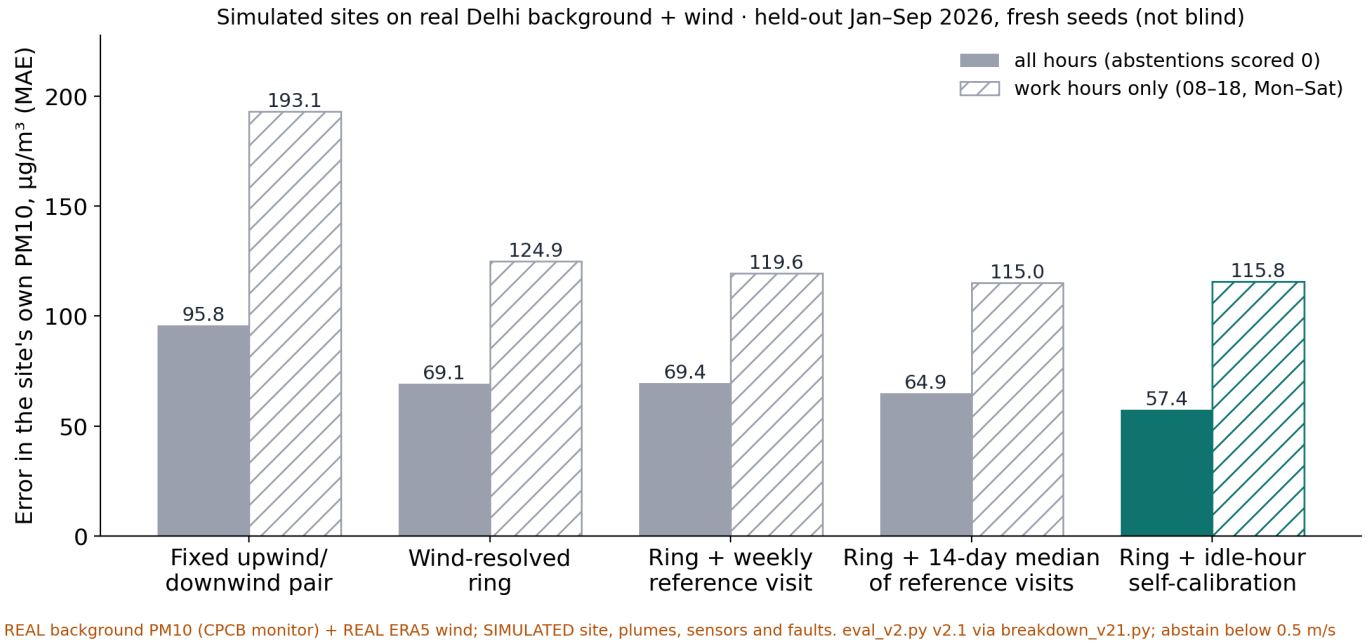


Upwind nodes breathe Delhi.
Downwind nodes breathe Delhi + the site.
The difference is the site's own dust.

Status labels: BUILT · SIMULATED · PLANNED · every run number is rendered from committed results (eval/render_docs.py)

Attribution, stress-tested

SIMULATED sites · REAL background + wind



REAL background PM10 (CPCB monitor) + REAL ERA5 wind; SIMULATED site, plumes, sensors and faults. eval_v2.py v2.1 via breakdown_v21.py; abstain below 0.5 m/s

All hours: self-calibration 57.4 vs 64.9 µg/m³ for the strongest reference-visit baseline; coverage 0.95 ("inconclusive" below 0.5 m/s). **Work hours, when dust is emitted: no better (115.8 vs 115.0).** Its gain is fewer false alarms above 50 µg/m³ (FAR 0.145 vs 0.305). Simulation only; fresh seeds, not blind.

Cost and power — from vendor listings and datasheets

Kit parts: 4 nodes + wind + gateway + misting	**₹60,711** (listed retail parts incl. GST; excludes masts, cabling, data plan, installation, upkeep — not a build cost)
One node	₹10,032
Wind sensor	₹14,810 — 24% of the kit
Power per node (model estimate)	10.2 Wh/day from datasheet currents + stated assumptions
10 W panel in December (model estimate)	2.1× the daily load (monthly-mean sunlight; smog weeks are darker)
Battery with no sun (model estimate)	12.1 days

For scale: one construction-site violation costs ₹1 lakh to ₹2 lakh under CAQM's 2024 schedule as reported. Willingness to pay is untested.

Prices seen by the lead on robu.in / amazon.in, 4 Oct 2026 (deliverables/bom/bom.csv). Datasheets: Sensirion SPS30, Espressif ESP32-S3; irradiance: NASA POWER (fact rows F20–F22).

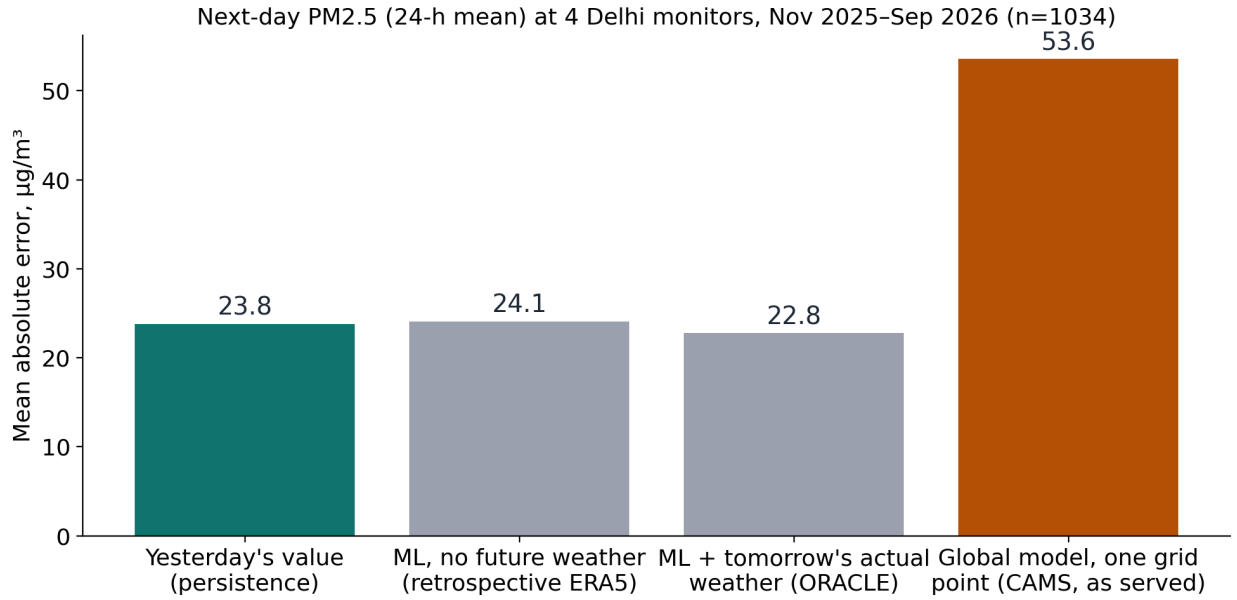
What we will test by 1 December — and our ask

When	What (approval-dependent)	Evidence it produces
5–19 Oct	order parts; build and bench-test one node, then four	measured current draw; working firmware
20–26 Oct	four nodes side by side; reference co-location if a slot is granted	inter-node agreement; calibration vs a reference
27–31 Oct	real-site trial **only if a site and approvals are in place** (Delhi's restrictions start 1 Nov)	first real fence-line estimates
Nov	approved, safety-controlled dust-source test: ring + wind + misting on/off in randomised blocks	estimate vs a known source; abstention rate; litres used
1 Dec	report + working prototype	real sensor data, labelled; never called field validation

Proposal, not a promise: an "earned-permission" pilot for verified-clean sites, for CAQM to consider.

Ask: a pilot site · a reference co-location slot · a DPCC/CAQM contact

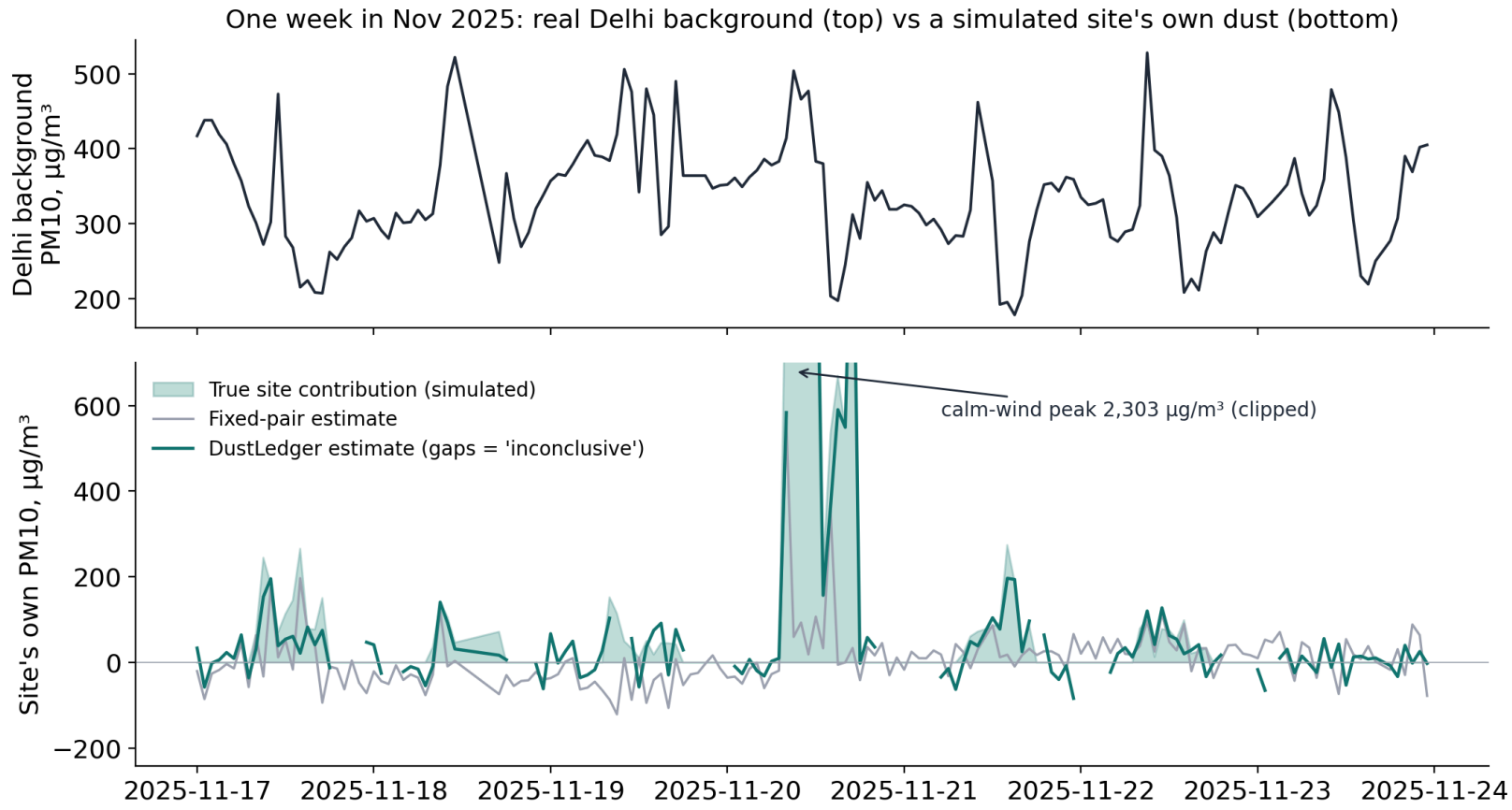
B1 · Forecasting on real Delhi data — why it isn't our spike BUILT



Real CPCB/DPCC data (OpenAQ archive) + ERA5. Walk-forward, time-ordered. eval/forecast_walkforward.py

Next-day PM2.5 at 4 CPCB/DPCC monitors (data from Feb 2025; walk-forward, n = 1,034 station-days): yesterday's value is off by 23.8 µg/m³. Our ML **without** future weather was −2.2% to −1.3% vs that; only an oracle with tomorrow's actual weather gained 4.4%.

B2 · One week of the replay SIMULATED site



Background = CPCB/DPCC monitor 6929 (real); wind = ERA5 (real); site + sensors SIMULATED (eval_v2.1, 'medium' scenario, seed 11).

B3 · What is built, simulated and planned

BUILT (committed, reproducible)	SIMULATED (labelled)	PLANNED (Round 2)
CPCB/DPCC hourly PM at 4 Delhi monitors (OpenAQ archive) + ERA5	construction sites, emissions, plumes	4-node ring, wind mast, gateway
forecast benchmark, walk-forward split	sensor gains, fouling, drift, faults	misting line with flow meter
attribution replay + frozen evaluators (v1, v2.1) + ledger	neighbour site, road, dirty nights	signed records + hash chain
typed contracts, verified parts list, power estimate, offline demo	misting effect: **not modelled**	swirling-wind check, supervisor app

B4 · Calm wind and "inconclusive"

Delhi, Nov–Jan (ERA5 grid, 2023–2026)	Share of hours
wind < 0.5 m/s (our current abstention threshold)	4.0%
wind < 1.0 m/s	16.2%
wind < 1.5 m/s	39.0%
median wind	1.7 m/s

ERA5 is a ~25 km grid. Calm at a real site, behind buildings and hoardings, has not been measured — that's one of the first things the ring will log. "Inconclusive" never counts as clean.

B5 · Stress-test details (evaluator v2.1, held-out Jan–Sep 2026)

MAE, µg/m³	Fixed pair	Ring	Ring + weekly ref. visit	Ring + 14-day ref. median	Ring + self-calibration
easy	80.1	43.6	41.3	38.6	36.5
medium	95.4	57.8	53.1	50.3	45.2
hard	101.9	81.8	86.0	77.5	66.7
hard, other site shape	105.9	93.3	97.2	93.2	81.3
work hours only	193.1	124.9	119.6	115.0	115.8

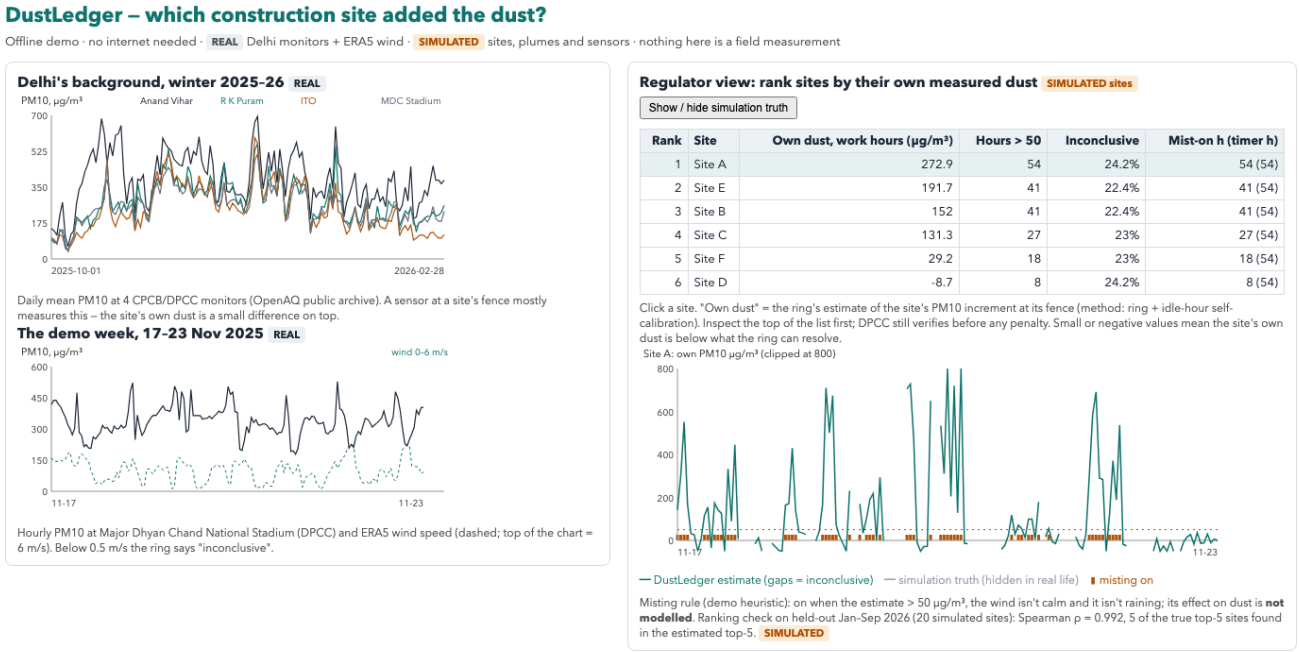
Truth = the site's contribution at its most-exposed node; "inconclusive" hours are scored as zero. Events above 50 µg/m³: self-calibration POD 0.676, FAR 0.145; 14-day reference median POD 0.707, FAR 0.305. Fresh seeds but **not blind**: the test period was seen during earlier versions, and the extra site shape is visible in the evaluator.

B6 · Parts list summary (prices seen 4 Oct 2026)

Subsystem	₹ (incl. GST)
4 nodes (SPS30, SHT30, ESP32-S3, SX1262, 10 W panel, LiFePO4 + MPPT charger, IP65 box)	40,128
Wind speed + direction sensor	14,810
Gateway (A7670C LTE, ESP32-S3, SX1262, box)	3,516
Misting kit + flow sensor + relay	2,257
Kit parts total (partial: not a build cost)	**60,711**

Not included: buck converters, cabling, masts, radiation shields, SIM/data, installation, reference co-location, upkeep. Every line with its URL: deliverables/bom/bom.csv. The relay listing was out of stock; the SX1262 module's IN865 band is unconfirmed.

B7 · Offline demo (screenshot) SIMULATED sites · REAL background + wind



Six simulated sites ranked by their own dust over a real Delhi week. Held-out check, 20 simulated sites: Spearman ρ 0.99, 5 of the true top-5 found — an easy test (emissions differ up to 30-fold).

B8 · Sources (selected)

- Delhi Winter Action Plan 2026-27 — The News Mill, Sept 2026 (reporting, not the order) · GRAP — Hindustan Times, Nov–Dec 2025
- CAQM penalty schedule (order of 6 Feb 2024) — Hindustan Times, Feb 2024 (reporting) · CAAQMS network — Hindustan Times, 2026
- Construction workers and aid — Down To Earth, Dec 2024 (survey by Aledia)
- Source apportionment — IIT Kanpur for DPCC, Jan 2016 (primary)
- Winter PM10 — TERI–ARAI (2016-17 campaign) · Low-cost sensors for construction dust — CSTEP guidance (2025)
- Datasheets — Sensirion SPS30 v0.9; Espressif ESP32-S3 v2.1 · Irradiance — NASA POWER climatology
- Data — CPCB/DPCC monitors via the OpenAQ archive; ERA5 and CAMS via Open-Meteo
- Every number with URL, quote and grade: research/fact-check.md · run numbers: eval/render_docs.py